Comparing models with ANOVA (answer key)

Each part has two scenarios with the same tasks. We'll work through scenario **A** together as a class. Then you'll work through scenario **B** in your group.

Every question in this handout compares a **simpler** model to a more **complex** model. Refer back to this box as you go.

$$F = \frac{(\text{RSS}_{simple} - \text{RSS}_{complex}) \ / \ (p_{complex} - p_{simple})}{\text{RSS}_{complex} \ / \ (n - p_{complex})}$$

- RSS: residual sum of squares, $\sum(y_i - \hat{y}_i)^2$. The variation the model leaves unexplained.
- p: number of coefficients the model estimates, including the intercept
- n: number of observations
- $df = n - p$: degrees of freedom left over after fitting the model
- **Numerator**: improvement in fit per df spent. **Denominator**: leftover noise per df remaining.

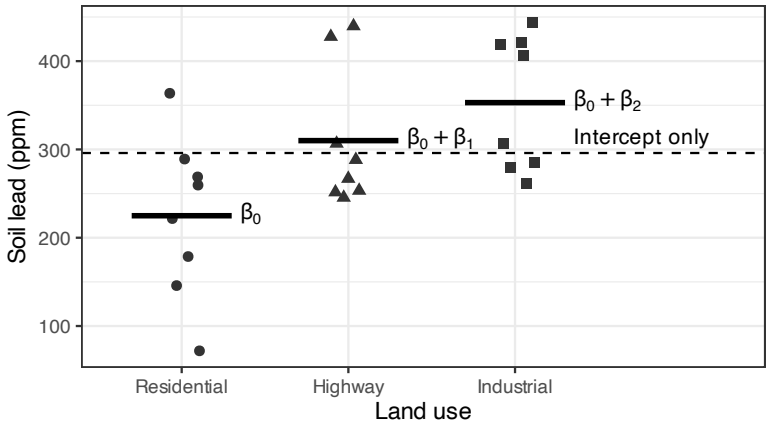
Part 1: Three levels

For each scenario:

1. **Data.** Look at the data in the figure.
2. **Dummy variables.** Fill in the dummy variable table.
3. **Full model.** Write the equation for the full model. What does each coefficient mean?
4. **Draw the means.** Use the coefficients to draw the intercept-only model and each category's mean on the figure.
5. **Degrees of freedom and F.** Use n and the RSS values to fill in p and df for each model. Then calculate F.
6. **Interpretation.** State H_0 and H_A. Then interpret the p-value for your F in a sentence.

1A. Soil lead (together)

Lead from old gasoline and industrial emissions settles into soil, where children can be exposed. A county health department sampled soil **lead** (ppm) in 8 yards near each of three land uses (**land_use**): residential streets, a highway, and an industrial site.



Dummy variables

Land use	Highway	Industrial
Residential	0	0
Highway	1	0
Industrial	0	1

Full model

$\text{lead} = \beta_0 + \beta_1(\text{land use} = \text{Highway}) + \beta_2(\text{land use} = \text{Industrial}) + \varepsilon$

β_0 is the mean soil lead in residential yards (the reference). β_1 is the difference between highway and residential yards. β_2 is the difference between industrial and residential yards.

Draw the means

```
coef(lm(lead ~ 1, yards))
```

```
(Intercept)
   295.9625
```

```
coef(lm(lead ~ land_use, yards))
```

```
      (Intercept)    land_useHighway land_useIndustrial
         224.9125            85.0625           128.0875
```

Intercept only: one line at the overall mean, 296.0 ppm. Residential: $\beta_0 = 224.9$. Highway: $\beta_0 + \beta_1 = 310.0$. Industrial: $\beta_0 + \beta_2 = 353.0$ ppm.

Degrees of freedom and F (n = 24)

Model	p	df	RSS
`lead ~ 1`	1	23	210,538.5
`lead ~ land_use`	3	21	142,556.7

$$F = \frac{(210,538.5 - 142,556.7) \;/\; (3-1)}{142,556.7 \;/\; (24-3)} = 5.01$$

Interpretation

State H_0 and H_A.

H_0: $\mu_{residential} = \mu_{highway} = \mu_{industrial}$ (equivalently, $\beta_1 = \beta_2 = 0$)

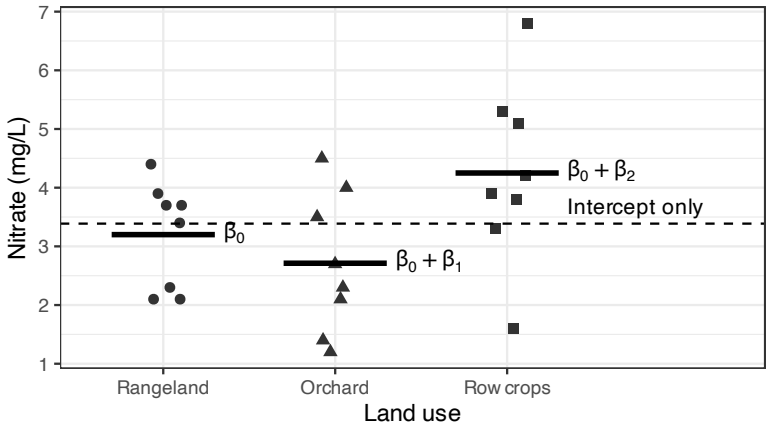
H_A: at least one land use has a different mean soil lead

The p-value for this F is p = 0.017. Interpret it in a sentence.

If land use had no relationship with soil lead, we'd see an F at least this large with probability p = 0.017. We reject H_0. Land use explains enough of the variation in soil lead to justify spending 2 df.

1B. Groundwater nitrate (in groups)

Fertilizer applied to crops can leach into groundwater as nitrate, contaminating drinking water wells. A water agency measured **nitrate** (mg/L) in 8 wells under each of three land uses (**land_use**): rangeland, orchards, and row crops.



Dummy variables

Land use	Orchard	Row crops
Rangeland	0	0
Orchard	1	0
Row crops	0	1

Full model

$\text{nitrate} = \beta_0 + \beta_1(\text{land use} = \text{Orchard}) + \beta_2(\text{land use} = \text{Row crops}) + \varepsilon$

β_0 is the mean nitrate under rangeland (the reference). β_1 is the difference between orchards and rangeland. β_2 is the difference between row crops and rangeland.

Draw the means

```
coef(lm(nitrate ~ 1, wells))
```

```
(Intercept)
     3.3875
```

```
coef(lm(nitrate ~ land_use, wells))
```

```
      (Intercept)   land_useOrchard land_useRow crops
           3.2000           -0.4875            1.0500
```

Intercept only: one line at the overall mean, 3.4 mg/L. Rangeland: $\beta_0 = 3.2$. Orchard: $\beta_0 + \beta_1 = 2.7$. Row crops: $\beta_0 + \beta_2 = 4.2$ mg/L.

Degrees of freedom and F (n = 24)

Model	p	df	RSS
`nitrate ~ 1`	1	23	42.2
`nitrate ~ land_use`	3	21	32.3

$$F = \frac{(42.2 - 32.3) \,/\, (3 - 1)}{32.3 \,/\, (24 - 3)} = 3.21$$

Interpretation

State H_0 and H_A.

H_0: $\mu_{rangeland} = \mu_{orchard} = \mu_{row\ crops}$ (equivalently, $\beta_1 = \beta_2 = 0$)

H_A: at least one land use has a different mean nitrate

The p-value for this F is p = 0.061. Interpret it in a sentence.

If land use had no relationship with nitrate, we'd see an F at least this large with probability p = 0.061. We fail to reject H_0. Row crops look a little higher in the figure, but the differences among land uses are small relative to the variation among wells.

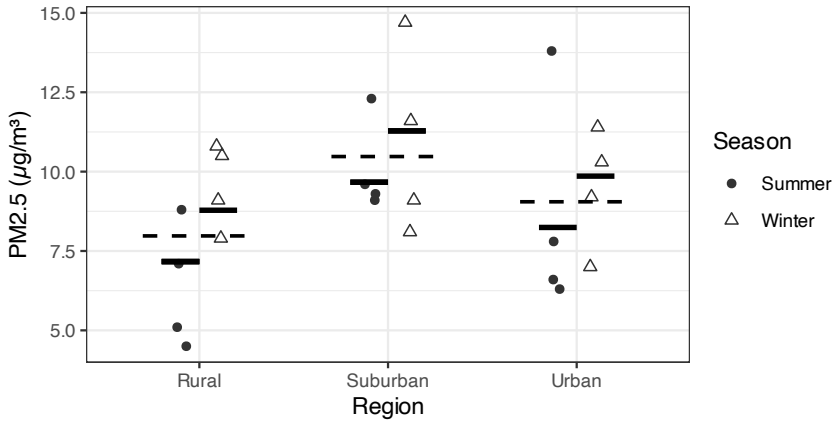
Part 2: Two categorical predictors

Now there are two categorical predictors. The question is whether the second one explains enough variation to justify its cost *after* accounting for the first. For each scenario:

1. **Data.** Look at the data in the figure.
2. **Full model.** Write the equation for the model with both predictors. What does each coefficient mean?
3. **Draw the models.** Use the coefficients to draw the simpler model's prediction for each level of the first predictor, and the model with both predictors' prediction for each combination of the two. Draw each prediction as a horizontal line segment, like in Part 1.
4. **Degrees of freedom and F.** Use n and the RSS values to fill in p and df for each model. Then calculate F for adding the second predictor.
5. **Interpretation.** State H_0 and H_A. Then interpret the p-value for your F in a sentence.

2A. Air quality (together)

Fine particulate matter (PM2.5) damages lungs and hearts. In winter, wood smoke and temperature inversions can trap it near the ground. An air district measured PM2.5 (**pm**, μg/m³) on 4 days for each combination of **region** (rural, suburban, or urban) and **season** (summer or winter). Does season matter after accounting for region?



Full model

$\text{pm} = \beta_0 + \beta_1(\text{region} = \text{Suburban}) + \beta_2(\text{region} = \text{Urban}) + \beta_3(\text{season} = \text{Winter}) + \varepsilon$

β_0 is the expected PM2.5 in rural regions in summer. β_1 and β_2 are the differences between suburban (or urban) and rural regions *in the same season*. β_3 is the difference between winter and summer, the same in every region.

Draw the models

```
coef(lm(pm ~ region, air))
```

```
   (Intercept) regionSuburban    regionUrban
         7.975          2.500          1.075
```

```
coef(lm(pm ~ region + season, air))
```

```
   (Intercept) regionSuburban    regionUrban   seasonWinter
      7.166667       2.500000       1.075000       1.616667
```

Simpler model (dashed): one prediction per region, Rural = 8.0; Suburban = 10.5; Urban = 9.1 μg/m³. Model with both: each prediction is β_0 plus the coefficients for its region and season, Rural, Summer = 7.2; Suburban, Summer = 9.7; Urban, Summer = 8.2; Rural, Winter = 8.8; Suburban, Winter = 11.3; Urban,

Winter = 9.9 μg/m³. In every region, the winter segment is β_3 above the summer segment.

Degrees of freedom and F (n = 24)

Model	p	df	RSS
`pm ~ region`	3	21	120.4
`pm ~ region + season`	4	20	104.7

Adding season costs 1 df.

$$F = \frac{(120.4 - 104.7) \;/\; (4 - 3)}{104.7 \;/\; (24 - 4)} = 2.99$$

Interpretation

State H_0 and H_A.

H_0: $\beta_3 = 0$ (season has no relationship with PM2.5 within each region)

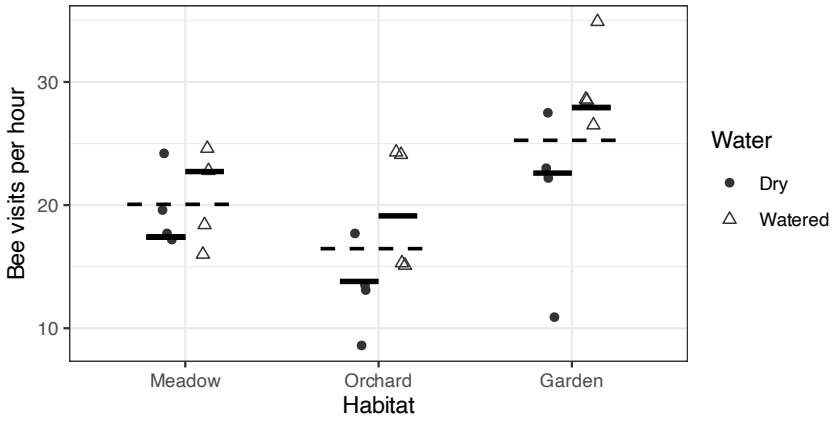
H_A: $\beta_3 \neq 0$

The p-value for this F is p = 0.099. Interpret it in a sentence.

If season had no relationship with PM2.5 within each region, we'd see an F at least this large with probability p = 0.099. We fail to reject H_0. Winter PM2.5 averaged 1.6 μg/m³ higher than summer, but that's small relative to the day-to-day variation within each region.

2B. Bees and watering (in groups)

Watered flowers may produce more nectar, attracting more bees. A pollinator team counted bee **visits** per hour to 4 flower patches for each combination of **habitat** (meadow, orchard, or garden) and **water** (dry or watered). Does watering matter after accounting for habitat?



Full model

visits $= \beta_0 + \beta_1(\text{habitat} = \text{Orchard}) + \beta_2(\text{habitat} = \text{Garden}) + \beta_3(\text{water} = \text{Watered}) + \varepsilon$

β_0 is the expected visits to dry meadow patches. β_1 and β_2 are the differences between orchard (or garden) and meadow patches *with the same watering*. β_3 is the difference between watered and dry patches, the same in every habitat.

Draw the models

```
coef(lm(visits ~ habitat, bees))
```

```
   (Intercept) habitatOrchard  habitatGarden 
       20.0625        -3.6000         5.2000 
```

```
coef(lm(visits ~ habitat + water, bees))
```

```
   (Intercept) habitatOrchard  habitatGarden   waterWatered 
        17.400         -3.600          5.200          5.325 
```

Simpler model (dashed): one prediction per habitat, Meadow = 20.1; Orchard = 16.5; Garden = 25.3 visits per hour. Model with both: each prediction is β_0 plus the coefficients for its habitat and watering, Meadow, Dry = 17.4; Orchard, Dry = 13.8; Garden, Dry = 22.6; Meadow, Watered = 22.7; Orchard, Watered = 19.1; Garden, Watered = 27.9 visits per hour. In every habitat, the watered segment is β_3 above the dry segment.

Degrees of freedom and F (n = 24)

Model	p	df	RSS
`visits ~ habitat`	3	21	626.7
`visits ~ habitat + water`	4	20	456.5

Adding watering costs 1 df.

$$F = \frac{(626.7 - 456.5) \,/\, (4 - 3)}{456.5 \,/\, (24 - 4)} = 7.45$$

Interpretation

State H_0 and H_A.

H_0: $\beta_3 = 0$ (watering has no relationship with bee visits within each habitat)

H_A: $\beta_3 \neq 0$

The p-value for this F is p = 0.013. Interpret it in a sentence.

If watering had no relationship with bee visits within each habitat, we'd see an F at least this large with probability p = 0.013. We reject H_0. Watered patches get 5.3 more visits per hour than dry patches in the same habitat.

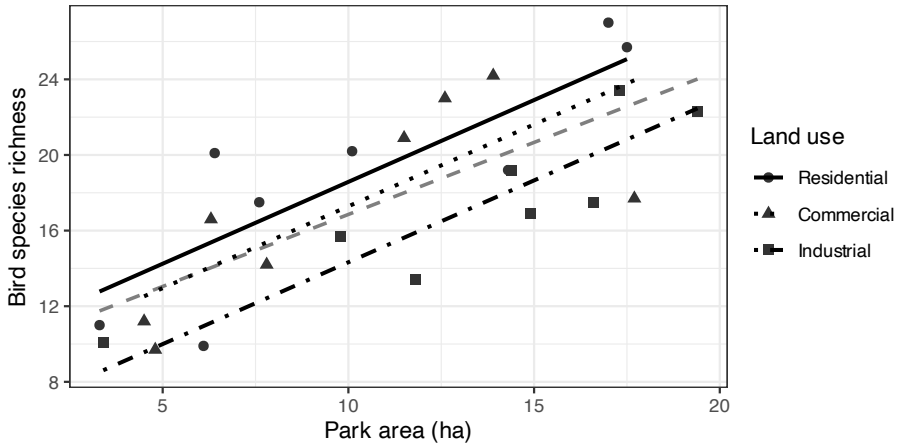
Part 3: Three levels plus a continuous predictor

Now the second predictor is continuous instead of categorical. The question is whether the categorical predictor explains enough variation to justify its cost *after* accounting for the continuous predictor. For each scenario:

1. **Full model.** Write the model equation with both predictors. What does each coefficient mean?
2. **Draw the models.** Use the coefficients to draw the model with only the continuous predictor (one line) and the model with both predictors (three parallel lines) on the figure.
3. **Degrees of freedom.** Fill in p and df for each model.
4. **F by hand.** Use the RSS values to calculate F for adding the categorical predictor.
5. **Check.** Compare your F to the `anova()` output.
6. **Conclusion.** State your conclusion in a sentence, using the p-value.

3A. Birds in city parks (together)

City parks can be refuges for birds, and bigger parks usually hold more species. Does what surrounds a park matter too? A city surveyed bird species **richness** in 8 parks in each of three surrounding land uses (**land_use**): residential, commercial, and industrial. It also recorded each park's **area** (ha).



Full model

richness $= \beta_0 + \beta_1$area $+ \beta_2$(land use $=$ Commercial) $+ \beta_3$(land use $=$ Industrial) $+ \varepsilon$

β_1 is the change in richness per hectare, the same slope for every land use. β_2 and β_3 are the differences between commercial (or industrial) parks and residential parks *of the same area*. β_0 is the expected richness of a residential park with area 0, which only sets the height of the lines.

Draw the models

```
coef(lm(richness ~ area, parks))
```

```
(Intercept)        area
  9.2357591   0.7618654
```

```
coef(lm(richness ~ area + land_use, parks))
```

```
      (Intercept)              area land_useCommercial land_useIndustrial
        9.9183439         0.8657746         -1.2911902         -4.2505121
```

Model with only area (dashed): intercept 9.2 and slope 0.76. Model with both: every line has slope $\beta_1 = 0.87$. The intercepts are Residential: $\beta_0 = 9.9$; Commercial: $\beta_0 + \beta_2 = 8.6$; Industrial: $\beta_0 + \beta_3 = 5.7$.

Degrees of freedom (n = 24)

Model	p	df	RSS
`richness ~ area`	2	22	257.9
`richness ~ area + land_use`	4	20	188.4

F by hand

Adding land use costs 2 df (two dummy variables).

$$F = \frac{(257.9 - 188.4) \;/\; (4 - 2)}{188.4 \;/\; (24 - 4)} = 3.69$$

Check

```
anova(lm(richness ~ area, parks), lm(richness ~ area + land_use, parks))
```

```
Analysis of Variance Table

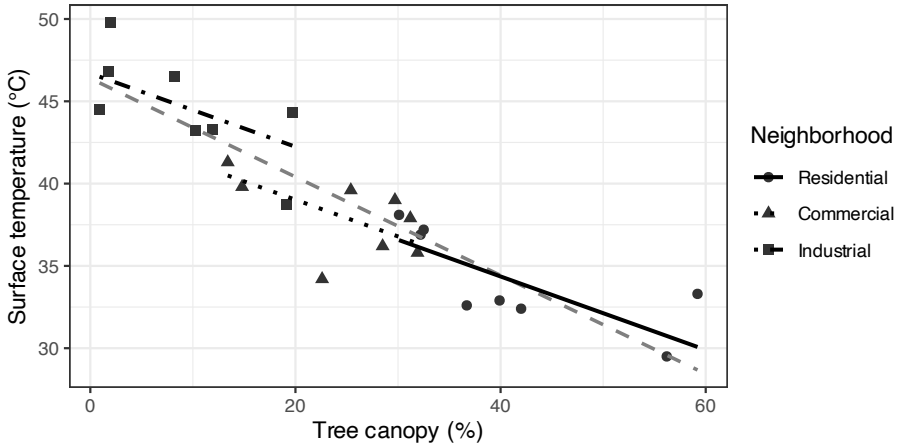
Model 1: richness ~ area
Model 2: richness ~ area + land_use
  Res.Df    RSS Df Sum of Sq      F  Pr(>F)
1     22 257.92
2     20 188.44  2    69.475 3.6867 0.04335 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusion

If land use had no relationship with richness among parks of the same size, we'd see an F at least this large with probability p = 0.043. We reject H_0. Compared to residential parks of the same area, commercial parks have 1.3 fewer species and industrial parks have 4.3 fewer.

3B. Urban heat (in groups)

Pavement and rooftops soak up summer sun, while trees shade the ground. A city measured summer surface temperature (**temp**, °C) at 8 sites in each of three **neighborhood** types: residential, commercial, and industrial. It also recorded the tree **canopy** cover (%) around each site. Does neighborhood type matter *after* accounting for canopy?



Full model

$$\text{temp} = \beta_0 + \beta_1 \text{canopy} + \beta_2(\text{neighborhood} = \text{Commercial}) + \beta_3(\text{neighborhood} = \text{Industrial}) + \varepsilon$$

β_1 is the change in temperature per percentage point of canopy, the same slope in every neighborhood. β_2 and β_3 are the differences between commercial (or industrial) and residential sites *with the same canopy*. β_0 is the expected temperature of a residential site with no canopy.

Draw the models

```
coef(lm(temp ~ canopy, sites))
```

```
(Intercept)      canopy
 46.3901866  -0.2991744
```

```
coef(lm(temp ~ canopy + neighborhood, sites))
```

```
          (Intercept)                canopy neighborhoodCommercial neighborhoodIndustrial
           43.2780196            -0.2230053              0.2024248              3.4194922
```

Model with only canopy (dashed): intercept 46.4 and slope -0.30. Model with both: every line has slope $\beta_1 = -0.22$. The intercepts are Residential: $\beta_0 = 43.3$; Commercial: $\beta_0 + \beta_2 = 43.5$; Industrial: $\beta_0 + \beta_3 = 46.7$.

Degrees of freedom (n = 24)

Model	p	df	RSS
`temp ~ canopy`	2	22	122.1
`temp ~ canopy + neighborhood`	4	20	96.1

F by hand

Adding neighborhood costs 2 df.

$$F = \frac{(122.1 - 96.1) \,/\, (4 - 2)}{96.1 \,/\, (24 - 4)} = 2.70$$

Check

```
anova(lm(temp ~ canopy, sites), lm(temp ~ canopy + neighborhood, sites))
```

```
Analysis of Variance Table

Model 1: temp ~ canopy
Model 2: temp ~ canopy + neighborhood
  Res.Df     RSS Df Sum of Sq     F  Pr(>F)
1     22 122.065
2     20  96.076  2    25.988 2.705 0.09126 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusion

If neighborhood type had no relationship with temperature among sites with the same canopy, we'd see an F at least this large with probability p = 0.091. We fail to reject H_0. Industrial sites are hotter in the figure, but they also have the least canopy. Once canopy is in the model, the evidence that neighborhood type matters is weak.